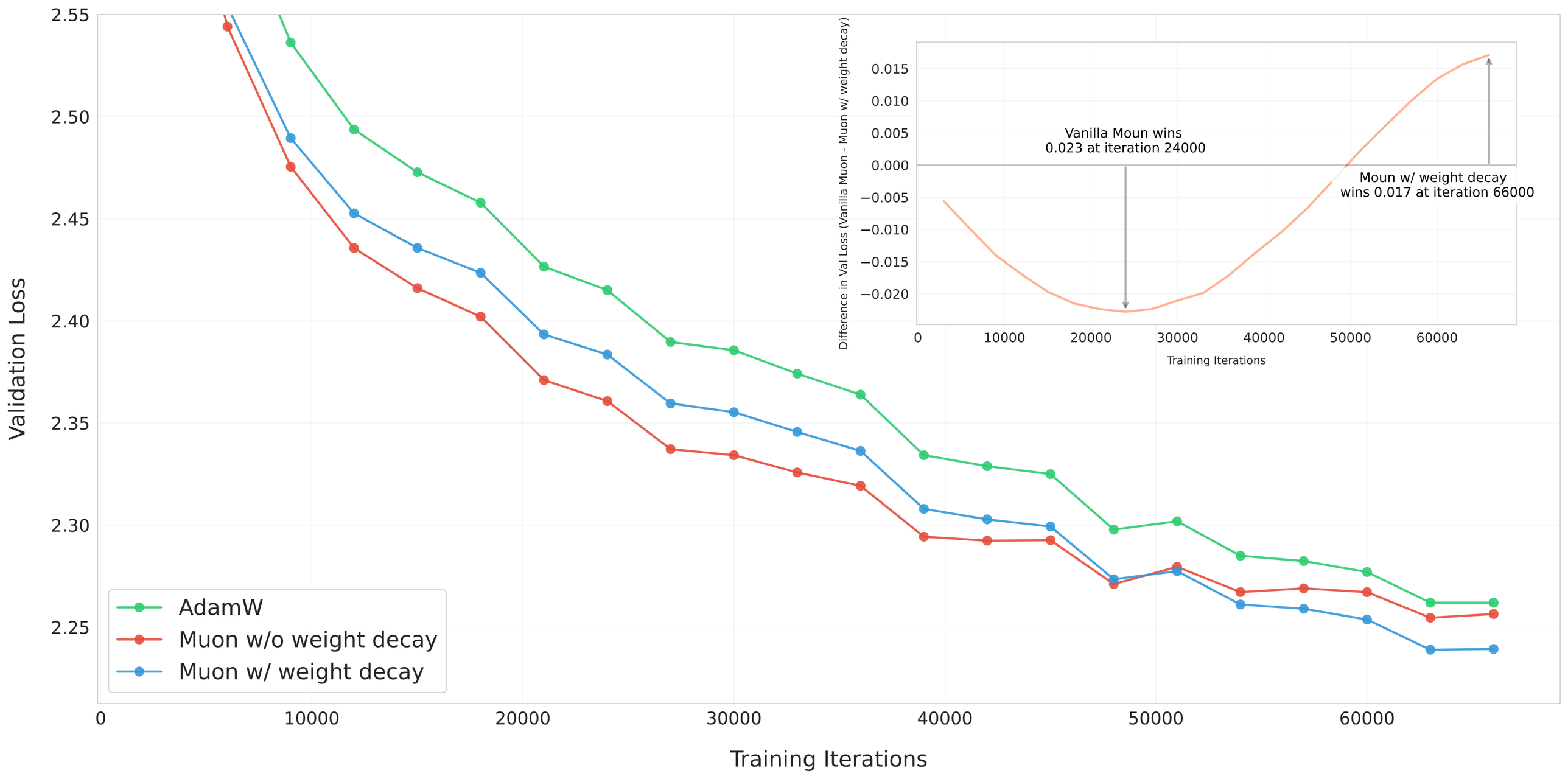
Validation Loss
2.55
2.50
2.45
2.40
2.35
2.30
2.25
0
10000
20000
30000
40000
50000
60000
Training Iterations
AdamW
Muon w/o weight decay
Muon w/ weight decay
Difference in Val Loss (Vanilla Muon - Muon w/ weight decay)
0.015
0.010
0.005
0.000
−0.005
−0.010
−0.015
−0.020
Vanilla Moun wins
0.023 at iteration 24000
Moun w/ weight decay
wins 0.017 at iteration 66000
0
10000
20000
30000
40000
50000
60000
Training Iterations